

Dr. Qusay Idrees Sarhan (Assistant Professor)



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CAREER OBJECTIVE

To be involved in teaching and research activities where I can utilize my skills and creativity to effectively contribute to the growth and development of the university I work at and myself.

ACADEMIC ACHIEVEMENTS

- | | |
|--------------------|---|
| 2020 - 2023 | PhD in Software Engineering (Scholarship), First Class with Distinction, University of Szeged, Szeged, Hungary. |
| 2009 - 2011 | M.Tech. in Software Engineering (Scholarship), First Class with Distinction, Jawaharlal Nehru Technological University, Telangana, India. |
| 2003 - 2007 | B.Sc. in Software Engineering, First Class with Distinction, University of Mosul, Nineveh, Iraq. |

AWARDS AND GRANTS

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|--------------------|---|
| 2023 - 2024 | A Prize for Earning PhD Early (graduate before time), by University of Szeged, Szeged, Hungary. |
| 2022 - 2023 | COST Visiting Grant Program, by the European Cooperation in Science and Technology (COST), Brussels, Belgium. |
| 2022 - 2023 | PhD Student's Excellence Scholarship, by University of Szeged, Szeged, Hungary. |
| 2020 - 2023 | Grant by the European Union project RRF-2.3.1-21-2022-00004 within the framework of the Artificial Intelligence National Laboratory. |
| 2020 - 2023 | Grant by the Ministry of Innovation and Technology of Hungary from the National Research, Development and Innovation Fund, financed under the TKP2021-NVA funding scheme. |
| 2020 - 2023 | Stipendium Hungaricum Scholarship for PhD's Degree, by the Hungarian Government, Budapest, Hungary. |
| 2020 - 2021 | Partial fund by the Knowledge Foundation of Sweden (KKS) through the Synergy Project AIDA - A Holistic AI-driven Networking and Processing Framework for Industrial IoT (Rek:20200067). |

2018 - 2019	MENA Scholarship Programme (MSP), by the Netherlands government, Maastricht, Netherlands.
2016 - 2017	DAAD Visiting Grant Program, by German Academic Exchange Service (DAAD) Organization, Bonn, Germany.
2013 - 2014	Fulbright Visiting Scholar Program (Engineering Cohort), by U.S. Department of State's Bureau of Educational and Cultural Affairs and the Institute of International Education's Council for International Exchange of Scholars, Washington DC, USA.
2008 - 2011	Kurdistan Regional Government (KRG) Scholarship for Master's Degree, by the Ministry of Higher Education and Scientific Research, Kurdistan Region, Iraq.

RESEARCH INTERESTS

- Internet of Things
- Software Engineering
- Cybersecurity
- Generative AI
- Large Language Models

POSTGRADUATE SUPERVISION

- **Hekar Azwar Mohammed Salih**

Master Thesis Title: Evaluating Large Language Models (LLMs) for Static Code Analysis.

Graduation Year: 2026.

- **Sardar Kaukaz Jabrw**

Master Thesis Title: Evaluating Large Language Models (LLMs) for Code Generation for Internet of Things (IoT) Constrained Devices.

Graduation Year: 2026.

PUBLICATIONS

(i) Journal Papers

- [1] Sardar K. Jabrw and Qusay Idrees Sarhan. Evaluating Large Language Models for Arduino Code Generation. ARO-The Scientific Journal of Koya University, No. 1, Vol. 14, pp. 75-85, 2026.
- [2] Hekar A. Mohammed Salih and Qusay Idrees Sarhan. A Study of Large Language Models in Detecting Python Code Violations. ARO-The Scientific Journal of Koya University, No. 2, Vol. 13, pp. 215-225, 2025.
- [3] Sardar K. Jabrw and Qusay Idrees Sarhan. A Systematic Survey on Large Language Models for Code Generation. ARO-The Scientific Journal of Koya University, No. 2, Vol. 13, pp. 83-99, 2025.
- [4] Hekar A. Mohammed Salih and Qusay Idrees Sarhan. A Systematic Survey on Large Language Models for Static Code Analysis. ARO-The Scientific Journal of Koya University, No. 1, Vol. 13, pp. 251-265, 2025.
- [5] Hassan Bapeer Hassan and Qusay Idrees Sarhan. Evaluating Python Static Code Analysis Tools Using FAIR Principles. IEEE Access, 12, pp. 173647-173659, 2024.
- [6] Qusay Idrees Sarhan and Arpad Beszedes. A Survey of Challenges in Spectrum-Based Software Fault Localization. IEEE Access, 10, pp. 10618-10639, 2022.

- [7] Qusay Idrees Sarhan, Bestoun S. Ahmed, Miroslav Bures, Kamal Z. Zamli. Software Module Clustering: An In-Depth Literature Analysis. *IEEE Transactions on Software Engineering*, 48(6), pp. 1905-1928, 2022.
- [8] Hassan B. Hassan, Saman A. Barakat, Qusay Idrees Sarhan. Survey on Serverless Computing. *Journal of Cloud Computing*, 10(39), pp. 1-29, 2021.
- [9] Qusay Idrees Sarhan. Performance Evaluation of Java Programming Strategies. *Academic Journal of Nawroz University (AJNU)*, 10(4), pp. 146-159, 2021.
- [10] Qusay Idrees Sarhan. Systematic Survey on Smart Home Safety and Security Systems using the Arduino Platform. *IEEE Access*, 8, pp. 128362-128384, 2020.
- [11] Qusay Idrees Sarhan. Best Practices and Recommendations for Writing Good Software. *Journal of Duhok University*, 22(1), pp. 90-105, 2019.
- [12] Qusay Idrees Sarhan. Internet of Things: A Survey of Challenges and Issues. *International Journal of Internet of Things and Cyber-Assurance*, 1(1), pp. 40-75, 2018.
- [13] Saman A. Barakat, Qusay Idrees Sarhan. Performance evaluation of list iteration methods in Java: an empirical study. *Innovaciencia*, 6(1), pp. 1-6, 2018.
- [14] Hassan B. Hassan, Qusay Idrees Sarhan. Performance evaluation of relational embedded databases: an empirical study. *Innovaciencia*, 6(1), pp. 1-9, 2018.
- [15] Qusay Idrees Sarhan, Idrees S. Gawdan. Web Applications and Web Services: A Comparative Study. *Science Journal of University of Zakho*, 6(1), pp. 35-41, 2018.
- [16] Qusay Idrees Sarhan, Idrees S. Gawdan. Java Message Service Based Performance Comparison of Apache ActiveMQ and Apache Apollo Brokers. *Science Journal of University of Zakho*, 5(4), pp. 307-312, 2017.
- [17] Idrees S. Kocher, Qusay Idrees Sarhan. Classifying Routing Algorithms upon Clustered Based Wireless Sensor Networks: A Survey. *ZANCO Journal of Pure and Applied Sciences*, 29(4), pp. 25-36, 2017.
- [18] Qusay Idrees Sarhan, Nechirvan A. Mohammed. Arduino Based Real-Time Home Control System: Design and Implementation. *Journal of Duhok University*, 19(1), pp. 133-142, 2016.
- [19] Idrees S. Gawdan, Qusay Idrees Sarhan. Performance Evaluation of Novel Secure Key Management Scheme Over BAN Wireless Sensor Networks. *Journal of Duhok University*, 19(1), pp. 179-188, 2016.
- [20] Idrees Sarhan Gawdan, Chee-Onn Chow, Tanveer A Zia, Qusay Idrees Sarhan. Cross-layer Based Security Solutions for Wireless Sensor Networks. *International Journal of Physical Sciences*, 6(17), pp. 4245-4254, 2011.

(ii) Conference Papers

- [21] Attila Szatmári, Qusay Idrees Sarhan, Peter Attila Soha, Gergo Balogh, Arpad Beszedes. On the Integration of Spectrum-Based Fault Localization Tools into IDEs. *Proceedings of the 1st ACM/IEEE Workshop on Integrated Development Environments (IDE)*, pp. 24-29, 2024.
- [22] Qusay Idrees Sarhan, Arpad Beszedes. Experimental Evaluation of a New Ranking Formula for Spectrum based Fault Localization. In *Proceedings of the 22nd International Working Conference on Source Code Analysis and Manipulation (SCAM)*, IEEE, pp. 276-280, 2023.
- [23] Qusay Idrees Sarhan, Tamas Gergely, Arpad Beszedes. Systematically Generated Formulas for Spectrum-Based Fault Localization. *IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*, pp. 344-352, 2023.

- [24] Qusay Idrees Sarhan, Hassan Bapeer Hassan, Arpad Beszedes. SFLaaS: Software Fault Localization as a Service. IEEE Conference on Software Testing, Verification and Validation (ICST), pp. 467-469, 2023.
- [25] Qusay Idrees Sarhan, Hassan Bapeer Hassan, Arpad Beszedes. Poster: Software Fault Localization as a Service (SFLaaS). IEEE Conference on Software Testing, Verification and Validation (ICST), pp. 482-485, 2023.
- [26] Qusay Idrees Sarhan, Arpad Beszedes. Poster: Improving Spectrum Based Fault Localization For Python Programs Using Weighted Code Elements. IEEE Conference on Software Testing, Verification and Validation (ICST), pp. 478-481, 2023.
- [27] Qusay Idrees Sarhan, Tamas Gergely, Arpad Beszedes. New Ranking Formulas to Improve Spectrum Based Fault Localization Via Systematic Search. In Proceedings of the International Conference on Software Testing, Verification and Validation Workshops (ICSTW), IEEE, pp. 306-309, 2022.
- [28] Qusay Idrees Sarhan. Enhancing Spectrum based Fault Localization Via Emphasizing Its Formulas with Importance Weight. In Proceedings of the 3rd International Workshop on Automated Program Repair (APR), ACM, pp. 53–60, 2022.
- [29] Qusay Idrees Sarhan, Arpad Beszedes. Effective Spectrum Based Fault Localization Using Contextual Based Importance Weight. In Proceedings of the International Conference on the Quality of Information and Communications Technology (QUATIC), Springer, pp. 93–107, 2022.
- [30] Attila Szatmari, Qusay Idrees Sarhan, Arpad Beszedes. Interactive fault localization for Python with CharmFL. In Proceedings of the 13th International Workshop on Automating Test Case Design, Selection and Evaluation (A-TEST), ACM, pp. 33–36, 2022.
- [31] Ferenc Horváth, Gergő Balogh, Attila Szatmári, Qusay Idrees Sarhan, Béla Vancsics, Árpád Beszédes. Interacting with interactive fault localization tools. In Proceedings of the 13th International Workshop on Automating Test Case Design, Selection and Evaluation (A-TEST), ACM, pp. 61–63, 2022.
- [32] Jawaher Abdulwahab Fadhil, Qusay Idrees Sarhan. A Survey on Internet of Things (IoT) Testing. In Proceedings of the 2nd International Conference on Computer Science and Software Engineering (CSASE), IEEE, pp. 77-83, 2022.
- [33] Qusay Idrees Sarhan, Bela Vancsics, Arpad Beszedes. Method Calls Frequency-Based Tie-Breaking Strategy for Software Fault Localization. In Proceedings of the 21st International Working Conference on Source Code Analysis and Manipulation (SCAM), IEEE, pp. 103-113, 2021.
- [34] Qusay Idrees Sarhan, Attila Szatmari, Rajmond Toth, Arpad Beszedes. CharmFL: A Fault Localization Tool for Python. In Proceedings of the 21st International Working Conference on Source Code Analysis and Manipulation (SCAM), IEEE, pp. 114-119, 2021.
- [35] Jawaher Abdulwahab Fadhil, Qusay Idrees Sarhan. Internet of Vehicles (IoV): A Survey of Challenges and Solutions. In Proceedings of the 21st International Arab Conference on Information Technology (ACIT), IEEE, pp. 1-10, 2021.
- [36] Qusay Idrees Sarhan. Arduino Based Smart Home Warning System. In Proceedings of the 6th International Conference on Control Science and Systems Engineering (ICCSSE), IEEE, pp. 201-206, 2020.
- [37] Jawaher Abdulwahab Fadhil, Omar Ammar Omar, Qusay Idrees Sarhan. A Survey on the Applications of Smart Home Systems. In Proceedings of the 1st International Conference on Computer Science and Software Engineering (CSASE), IEEE, pp. 168-173, 2020.
- [38] Hassan Bapeer Hassan, Qusay Idrees Sarhan. Performance Evaluation of Graphical User Interfaces in Java and C#. In Proceedings of the 1st International Conference on Computer Science and Software Engineering (CSASE), IEEE, pp. 290-295, 2020.

- [39] Qusay Idrees Sarhan, Roman Eppinger, Stefan Graf, Martin Nguyen, Thomas Schanze. Design and Implementation of a Teaching System for Visual Stimulation and Recording of Single Unit and Mass Signals. In Proceedings of the Annual Meeting of the German Society of Biomedical Engineering and Joint Conference in Medical Physics (BMTMedPhys), pp. 67-67, 2017.
- [40] Idrees Sarhan Gawdan, Chee-Onn Chow, Tanveer A Zia, Qusay Idrees Sarhan. A Novel Secure Key Management Module for Hierarchical Clustering Wireless Sensor Networks. In Proceedings of the 3rd International Conference on Computational Intelligence, Modelling and Simulation (CIMSIM), IEEE, pp. 312-316, 2011.

PAPER REVIEW SERVICES

I have reviewed research papers for the following journals and conferences:

- IEEE Transactions on Software Engineering
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Emerging Topics in Computing
- IEEE Internet of Things
- IEEE Software
- IEEE Access
- IEEE IT Professional
- ACM Transactions on Internet Technology
- ACM Transactions on Software Engineering and Methodology
- ACM Transactions on Intelligent Systems and Technology
- ACM Transactions on Design Automation of Electronic Systems
- ACM Computing Surveys
- Expert Systems with Applications
- The Journal of Supercomputing
- Cluster Computing: The Journal of Networks, Software Tools and Applications
- World Wide Web: Internet and Web Information Systems (WWW)
- BMC Public Health
- Journal of Testing and Evaluation
- International Journal of Intelligent Systems
- IET Software
- Computing
- Journal of Data and Information Quality
- Computers, Materials & Continua
- PeerJ Computer Science
- PLoS ONE
- ARO-The Scientific Journal of Koya University
- Mathematical Problems in Engineering
- Journal of Control Science and Engineering
- Diyala Journal of Engineering Sciences
- International Conference on Human Factors in Computing Systems (CHI)
- International Conference on Computer Science and Software Engineering (CSASE)
- International Conference on Advanced Science and Engineering (ICOASE)
- International Conference on Recent Innovations in Engineering (ICRIE)

WORK EXPERIENCES

- 2012 - present** University faculty member, department of computer science, college of science, University of Duhok.
- 2020 - 2023** Researcher, department of Software Engineering, institute of informatics, University of Szeged, Szeged, Hungary.
Project Title: Methods for Enhancing Software Fault Localization.
Period: 3 years.
- 2017 - 2018** Trainer, Voice of Older People and Family (VOP-Fam) Organization, Duhok, Kurdistan Region, Iraq.
Project Title: Computer Basics as a part of “Adolescent and Youth Development in Five IDPs Camps in Duhok” project.
Period: 1 month.
- 2016 - 2017** Visiting scholar, department of biomedical engineering, college of engineering, Technische Hochschule Mittelhessen (THM) University of Applied Sciences, Giessen, Germany.
Project Title: Design and Implementation of a Teaching System for Visual Stimulation and Recording of Single Unit and Mass Signals
Period: 6 months.
- 2013 - 2014** Visiting scholar, department of computer and information sciences, college of engineering, University of Delaware, Delaware, USA.
Project Title: Knowledge and Capacity Building on Best Practices in Engineering.
Period: 2.5 months.
- 2008 - 2011** Researcher, department of Software Engineering, school of IT, Jawaharlal Nehru Technological University, Telangana, India.
Project Title: Vision Based Home Monitoring System Using Java Technologies.
Period: 2 years.
- 2007 - 2008** Teaching assistant, department of electrical and computer engineering, college of engineering, University of Duhok.
Period: 6 months.

TEACHING EXPERIENCES

- 2023 - present** Internet of Things (IoT), bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2023 - present** Java Programming, bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2018 - 2019** Computer Skills, bachelor level course, business administration dept., University of Nawroz, Duhok, Kurdistan Region, Iraq.
- 2017 - 2019** Visual Programming, diploma level course, computer systems dept., Ararat Private Technical Institute, Duhok, Kurdistan Region, Iraq.
- 2017 - 2019** C++ Programming, diploma level course, computer systems dept., Ararat Private

Technical Institute, Duhok, Kurdistan Region, Iraq.

- 2017 - 2018** Systems Analysis, diploma level course, information technology dept., Duhok Private Technical Institute, Duhok, Kurdistan Region, Iraq.
- 2014 - 2020** Embedded Systems, bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2014 - 2015** Linux Programming, bachelor level course, electrical and computer engineering dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2012 - 2016** Linux Programming, bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2012 - 2020** Java Technology, bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2011 - 2012** Scientific Debates, bachelor level course, computer science dept., University of Duhok, Duhok, Kurdistan Region, Iraq.
- 2007 - 2008** C++ Programming, bachelor level lab course, electrical and computer engineering dept., University of Duhok, Duhok, Kurdistan Region, Iraq.

TRAINING COURSES

- 2022 - 2023** “Privacy-Aware Audio- and Video-Based Applications for Active and Assisted Living (AAL)”, 6-day training, European Cooperation in Science and Technology (COST), Skopje, Macedonia.
- 2018 - 2019** “Digital Governments and Big Data”, 12-day training, Maastricht School of Management (MSM), Maastricht, Netherlands.
- 2016 - 2017** “Software Development for BioMedical Engineering”, 6-month training, Technische Hochschule Mittelhessen (THM) University of Applied Sciences, Giessen, Germany.
- 2015 - 2016** “Cultural Identity and Intercultural Competence”, 1-day training, Technische Hochschule Mittelhessen (THM) University of Applied Sciences, Giessen, Germany.
- 2013 - 2014** “Science Visualization”, 2-day training, University of Delaware, Delaware, USA.
- 2013 - 2014** “Software Development for Mechatronics and Embedded Systems”, 2.5-month training, University of Delaware, Delaware, USA.
- 2012 - 2013** “Teaching Methods for Higher Education”, 2-month training, University of Duhok, Kurdistan Region, Iraq.
- 2009 - 2010** “Software Development for Wireless Sensor Networks”, 20-day training, University of Malaya, Kuala Lumpur, Malaysia.

CO-CURRICULAR ACTIVITIES

(i) Seminars and Workshops

- Attended, presented, and participated in seminars and workshops in USA, Germany, Netherlands, Spain, Hungary, India, Malaysia, Macedonia, and Iraq.

(ii) Letters of Gratitude & Appreciation

2025 - 2026	Most Publications – for research productivity, Dean of College of Science, University of Duhok.
2025 - 2026	Outstanding Collaboration Recognition – for effective academic and research collaboration, Dean of College of Science, University of Duhok.
2025 - 2026	Research Funding Achievement – for successfully securing competitive research funding, Dean of College of Science, University of Duhok.
2025 - 2026	Publishing Papers in Q1 and Q2 journals, University President, University of Duhok.
2025 - 2026	Outstanding achievements and services, University President, University of Duhok.
2024 - 2025	Publishing Papers in Q1 and Q2 journals, University President, University of Duhok.
2021 - 2022	Publishing Papers in Q1 and Q2 journals, University President, University of Duhok.
2018 - 2019	Scientific committee member of the 5 th annual Rwanga Awards, Rwanga organization, Duhok, Kurdistan region, Iraq.
2017 - 2018	Outstanding management member of the 6 th annual software exhibition, department of computer science, University of Duhok.
2016 - 2017	Outstanding achievements and services, University President, University of Duhok.
2016 - 2017	Outstanding management member of the 5 th annual software exhibition, department of computer science, University of Duhok.
2015 - 2016	Outstanding exams management member, department of computer science, University of Duhok.
2015 - 2016	Outstanding management member of the 4 th annual software exhibition, department of computer science, University of Duhok.
2014 - 2015	Outstanding faculty member, department of computer science, University of Duhok.
2014 - 2015	Outstanding management member of the 3 rd annual software exhibition, department of computer science, University of Duhok.
2014 - 2015	Linux operating system training course for the Ministry of Transportation and Communication, Duhok, Kurdistan Region, Iraq.
2013 - 2014	Outstanding achievement in Teaching of Quality Assurance (TQA), University of Duhok.
2011 - 2012	Outstanding exams management member, department of computer science, University of Duhok.

- 2008 - 2009** The most sincere and diligent student for the English Language Course, IEC College of Engineering and Technology, UP, India.
- 2006 - 2007** Outstanding bachelor graduation project, University of Mosul, Nineveh, Iraq.
- 2004 - 2005** Outstanding participation in scientific exhibitions, University of Mosul, Nineveh, Iraq.

(iii) Certificates

- 2017 - 2018** The 6th Software Exhibition, Certificate of Participation, University of Duhok.
- 2017 - 2018** Trainer at Voice of Older People and Family (VOP-Fam) Organization, Certificate of Participation, Duhok, Kurdistan Region, Iraq.
- 2016 - 2017** The 5th Software Exhibition, Certificate of Participation, University of Duhok.
- 2015 - 2016** International Buddy Program, Certificate of Completion, Technische Hochschule Mittelhessen (THM) University of Applied Sciences, Giessen, Germany.
- 2015 - 2016** The 4th Software Exhibition, Certificate of Participation, University of Duhok.
- 2013 - 2014** Fulbright Visiting Scholar Program, Certificate of Completion, University of Delaware, Delaware, USA.
- 2011 - 2012** Microsoft Imagine Cup, Certificate of Participation, IT academy, Erbil, Kurdistan Region, Iraq.
- 2008 - 2009** English Language Course, Certificate of Completion, IEC College of Engineering and Technology, UP, India.
- 2007 - 2008** English Language Proficiency Test, University of Duhok.
- 2007 - 2008** Computer Proficiency Test, University of Duhok.

PROFESSIONAL SKILLS

- Programming Languages : Python, MicroPython, Java, Arduino, C++
- Software Development Tools : PyCharm, NetBeans, Arduino IDE
- Software Testing Tools : Pytest, Coverage.py, Pylint, Radon, Bandit
- AI Development Tools : Ollama, LangChain, FAISS
- Operating Systems : Windows and Linux
- Research and Academic Tools : LaTeX and Mendeley

PROFESSIONAL INTERNATIONAL AFFILIATION/MEMBERSHIP

- 2023 - present** Szeged Alumni Network, Hungary.
- 2019 - present** NL Alumni Network, Netherlands.
- 2015 - present** USA International Exchange Alumni (IEA), USA.

SCIENTIFIC AND ADMINISTRATION COMMITTEES AT UNIVERSITY OF DUHOK

- Head of the Computer Science's Annual Software Fair Committee.
- Head of the Computer Science's Examinations Management Committee.
- Head of the Computer Science's Final Year Projects Evaluation Committee.
- Member of the Computer Science's Scientific Committee.
- Member of the Computer Science's Teaching of Quality Assurance Committee.

REFERENCES

- Provided on request